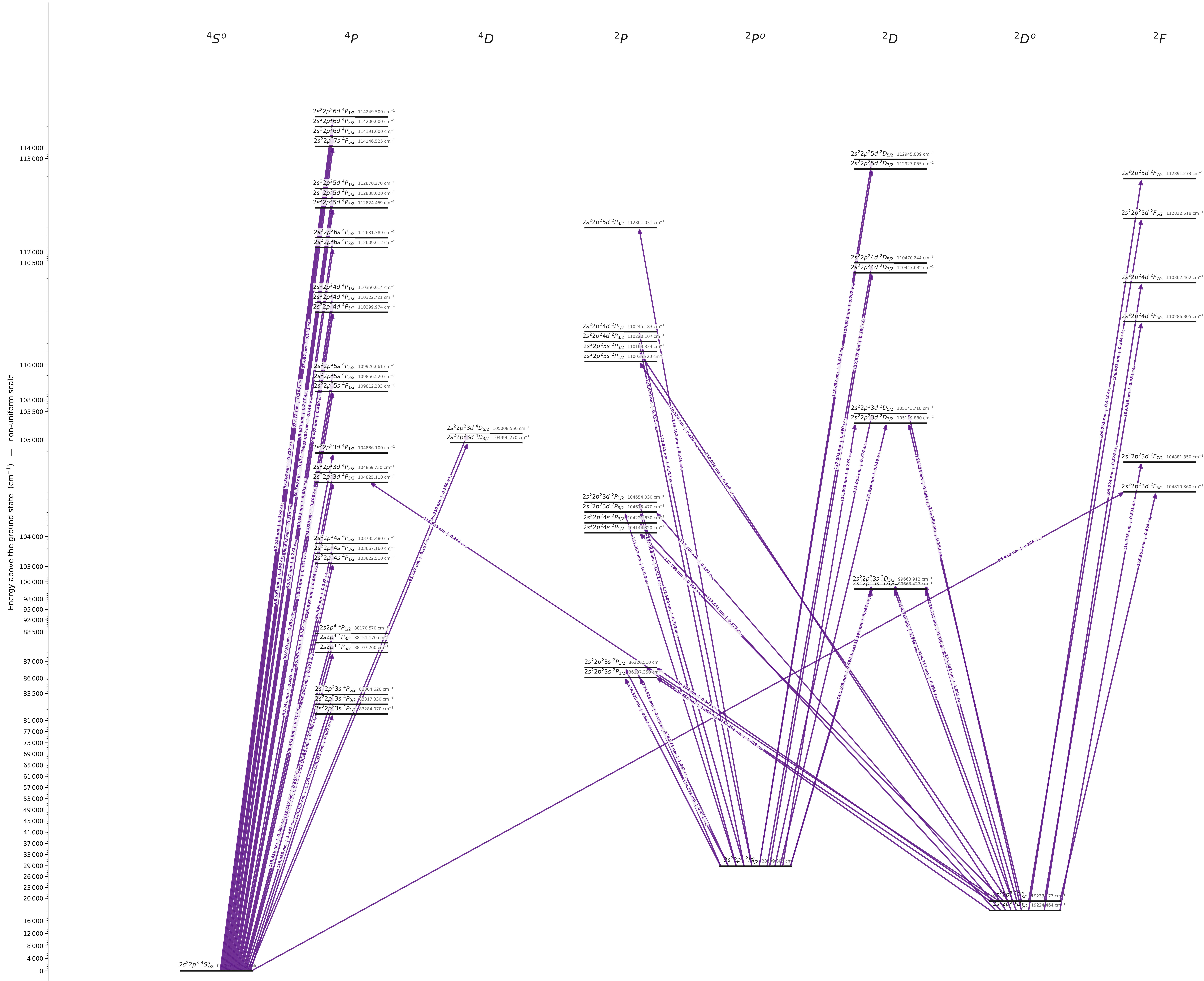
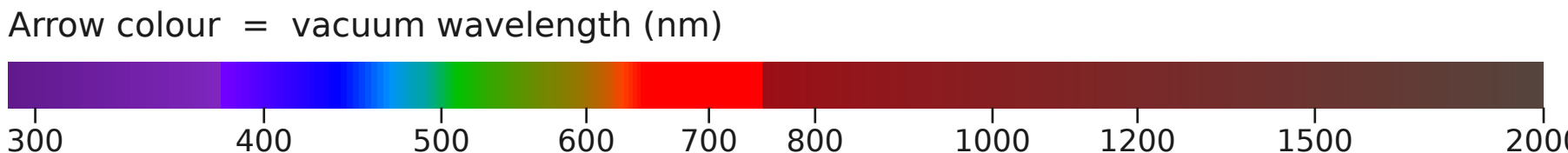


Nitrogen

N (N I). Strongest classified transitions below 2 μm (NIST): 59 levels, 70 lines.

N⁺ ionisation limit 117225.70 cm⁻¹ = 14.53413 eV (85.306 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $| \langle J || e r || J' \rangle |$ in *ea*₀
Matrix elements are measured or NIST values; * = high-accuracy theory, = = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.
Level caption: NIST energy and measured lifetime (where one exists).



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨J er J'⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
2s ² 2p ³ 4S _{3/2} – 2s ² 2p ² 3d ⁴ P _{1/2}	95.3415	–	3144.4062	0.403	1.900e+08	–	–	–
2s ² 2p ³ 4S _{3/2} – 2s ² 2p ² 3d ⁴ P _{3/2}	95.3655	–	3143.6156	0.557	1.810e+08	–	–	–
2s ² 2p ³ 4S _{3/2} – 2s ² 2p ² 3d ⁴ P _{5/2}	95.3970	–	3142.5777	0.645	1.620e+08	–	–	–
2s ² 2p ³ 4S _{3/2} – 2s2p ⁴ 4P _{1/2}	113.4165	–	2643.2872	0.466	1.510e+08	–	–	–
2s ² 2p ³ 4S _{3/2} – 2s2p ⁴ 4P _{3/2}	113.4415	–	2642.7056	0.655	1.490e+08	–	–	–
2s ² 2p ³ 4S _{3/2} – 2s2p ⁴ 4P _{5/2}	113.4980	–	2641.3892	0.790	1.440e+08	–	–	–
2s ² 2p ³ 2D _{3/2} – 2s ² 2p ² 3d ² F _{7/2}	116.7448	–	2567.9288	0.831	1.100e+08	–	–	–
2s ² 2p ³ 2D _{3/2} – 2s ² 2p ² 4s ² P _{1/2}	117.7695	–	2545.5870	0.407	1.030e+08	–	–	–
2s ² 2p ³ 4S _{3/2} – 2s ² 2p ² 3s ⁴ P _{3/2}	119.9550	–	2499.2084	1.442	4.070e+08	–	–	–
2s ² 2p ³ 4S _{3/2} – 2s ² 2p ² 3s ⁴ P _{5/2}	120.0223	–	2497.8057	1.173	4.030e+08	–	–	–
2s ² 2p ³ 4S _{3/2} – 2s ² 2p ² 3s ⁴ P _{1/2}	120.0710	–	2496.7936	0.827	4.000e+08	–	–	–
2s ² 2p ³ 2D _{3/2} – 2s ² 2p ² 3s ² D _{3/2}	124.3179	–	2411.4994	1.354	3.220e+08	–	–	–
2s ² 2p ³ 2D _{3/2} – 2s ² 2p ² 3s ² D _{5/2}	124.3306	–	2411.2528	1.085	3.100e+08	–	–	–
2s ² 2p ³ 2D _{3/2} – 2s ² 2p ² 3s ² P _{3/2}	149.2625	–	2008.4909	1.429	3.110e+08	–	–	–
2s ² 2p ³ 2D _{3/2} – 2s ² 2p ² 3s ² P _{1/2}	149.4675	–	2005.7367	1.068	3.460e+08	–	–	–
2s ² 2p ³ 2P _{3/2} – 2s ² 2p ² 3s ² P _{3/2}	174.2731	–	1720.2452	1.047	1.050e+08	–	–	–

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).